

final

Queueing System with Finite Population, Finite Buffer Size, and s -Servers

Consider a Queueing system that has **Finite population**, **Finite buffer size**, and the number of servers is **s -Servers**.

Specifically, assume:

- A finite population of M terminals
- Each terminal has an arrival rate of λ
- The system has s servers
- Each server has service parameter μ
- The system also has a finite queue size such that the **total number of customers in the system** (queueing plus those in service) is **no more than K**

That is:

- s customers can be in service
- $K - s$ customers can wait in the queue

Note that:

$$s \leq K \leq M$$

(a)

Draw the **state transition diagram** for this Queueing system and **clearly show the arrival and service rates** between states in your figure.

(b)

Solve for P_n in terms of:

$P_0, n, \lambda, M, s, \mu$ ONLY.

This is a **finite-source, finite-capacity multi-server queue**:

M/M/s/K/M

- M = Population
- s = Servers
- K = Capacity in system, $K - s$ = users in queue
- λ = Arrival rate (per user)
- μ = Service rate (per server)

s.t: $s \leq K \leq M$

- **Finite buffer** → system blocks at (K)

Let state $n = 0, 1, \dots, K$ = number of customers in system.

Arrival rates

Finite population → arrivals decrease as system fills. Only users **not in system** can generate arrivals:

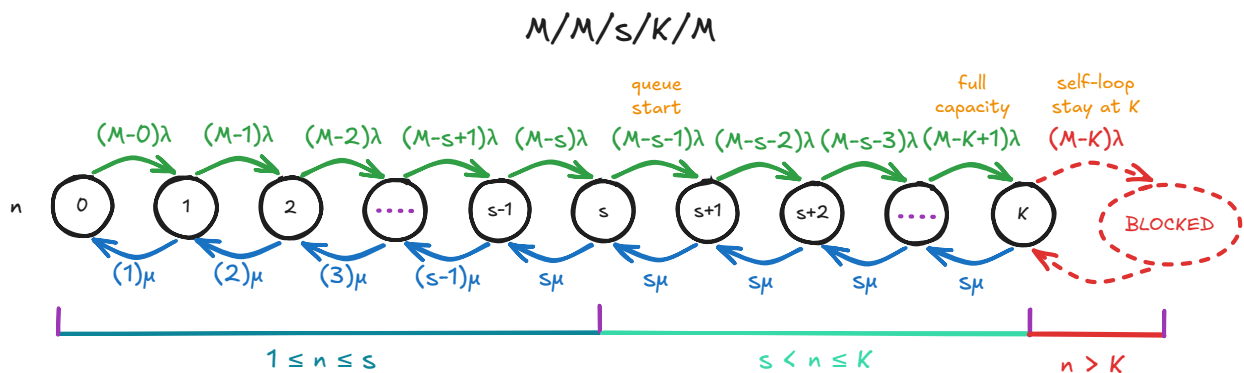
$$\lambda_n = (M - n)\lambda$$

Service rates

Finite Multi-server → service rate increases with arrivals then saturates at s , stays at s until n arrivals $< s$ again

$$\mu_n = \begin{cases} n\mu & n < s \\ s\mu & n \geq s \end{cases}$$

a) State Diagram:



b)

We define:

- P_n : probability that there are n customers in the system

Using balance equation:

$$M\lambda P_0 = \mu P_1 \rightarrow P_1 = \frac{M\lambda}{\mu} P_0$$

$$\therefore (M-1)\lambda P_1 = 2\mu P_2 \rightarrow P_2 = \frac{(M-1)\lambda}{2\mu} P_1$$

$$\therefore P_2 = \frac{(M-1)\lambda}{2\mu} \cdot \frac{M\lambda}{\mu} P_0$$

$$\therefore P_2 = \frac{M(M-1)}{2!} \left(\frac{\lambda}{\mu} \right)^2 P_0$$

$$\therefore (M-2)\lambda P_2 = 3\mu P_3 \rightarrow P_3 = \frac{(M-2)\lambda}{3\mu} P_2$$

$$\therefore P_3 = \frac{(M-2)\lambda}{3\mu} \cdot \frac{M(M-1)}{2!} \left(\frac{\lambda}{\mu} \right)^2 P_0$$

$$\therefore P_3 = \frac{M(M-1)(M-2)}{3!} \left(\frac{\lambda}{\mu} \right)^3 P_0$$

Continuing similarly:

$$\therefore P_n = \frac{M(M-1)(M-2)\cdots(M-n+1)}{n!} \left(\frac{\lambda}{\mu} \right)^n P_0$$

We notice the general pattern:

$$\therefore P_n = \frac{M!}{(M-n)!n!} \left(\frac{\lambda}{\mu} \right)^n P_0 \quad \forall n \in 0 \leq n \leq s \quad (1a)$$

 Recall the factorial definition:

$$M! = M(M-1)(M-2)\cdots(M-n+1)(M-n)(M-n-1)\cdots 1$$

If we divide by $(M-n)!$, all remaining lower terms cancel:

$$\frac{M!}{(M-n)!} = \frac{M(M-1)(M-2)\cdots(M-n+1)(M-n)(M-n-1)\cdots 1}{(M-n)(M-n-1)\cdots 1}$$

Everything after $(M-n+1)$ cancels, leaving:

$$\frac{M!}{(M-n)!} = M(M-1)(M-2)\cdots(M-n+1)$$

Binomial coefficient formula ("M choose n"):

$$\therefore \frac{M!}{(M-n)!n!} = \binom{M}{n}$$

$$\therefore P_n = \binom{M}{n} \left(\frac{\lambda}{\mu} \right)^n P_0 \quad \forall n \in 0 \leq n \leq s \quad (1b)$$

at $n = s$, equation becomes:

$$P_s = \frac{M!}{(M-s)!s!} \left(\frac{\lambda}{\mu} \right)^s P_0 \quad (2)$$

For $n > s$:

After state s , all servers are busy, so the service rate becomes constant
 $\mu_n = s\mu$

$$\therefore (M-s)\lambda P_s = s\mu P_{s+1} \rightarrow P_{s+1} = \frac{(M-s)\lambda}{s\mu} P_s$$

We sub P_s from equation 2:

$$\begin{aligned} \therefore P_{s+1} &= \frac{(M-s)\lambda}{s\mu} \cdot \frac{M!}{(M-s)!s!} \left(\frac{\lambda}{\mu}\right)^s P_0 \\ \therefore P_{s+1} &= \frac{M!}{s(M-s-1)!s!} \left(\frac{\lambda}{\mu}\right)^{s+1} P_0 \end{aligned}$$

$$\begin{aligned} \therefore (M-(s+1))\lambda P_{s+1} &= s\mu P_{s+2} \rightarrow P_{s+2} = \frac{(M-s-1)\lambda}{s\mu} P_{s+1} \\ \therefore P_{s+2} &= \frac{(M-s-1)\lambda}{s\mu} \cdot \frac{M!}{s(M-s-1)!s!} \left(\frac{\lambda}{\mu}\right)^{s+1} P_0 \\ \therefore P_{s+2} &= \frac{M!}{s^2(M-s-2)!s!} \left(\frac{\lambda}{\mu}\right)^{s+2} P_0 \end{aligned}$$

 Note the equations:

$$\frac{(M-s-1)}{(M-s-1)!} = \frac{\cancel{(M-s-1)}}{\cancel{(M-s-1)}(M-s-2)!} = \frac{1}{(M-s-2)!}$$

and

$$\lambda/\mu \cdot (\lambda/\mu)^{s+1} = (\lambda/\mu)^{s+2}$$

and

$$P_{n=s+2} \rightarrow \frac{\dots}{s^2 \dots} = \frac{\dots}{s^{(s+2)-s} \dots} = \frac{\dots}{s^{n-s} \dots}$$

We notice the general pattern:

$$P_n = \frac{M!}{s^{n-s} (M-n)!s!} \left(\frac{\lambda}{\mu}\right)^n P_0, \quad \forall n \in s < n \leq K \quad (3a)$$

if we want to put it in the binomial pattern similar to equation 1, we can multiply by $n!/n!$:

$$\begin{aligned} \therefore P_n &= \frac{n!}{s!s^{n-s}} \cdot \frac{M!}{(M-n)!n!} \left(\frac{\lambda}{\mu}\right)^n P_0, \\ \therefore P_n &= \frac{n!}{s!s^{n-s}} \binom{M}{n} \left(\frac{\lambda}{\mu}\right)^n P_0 \quad s < n \leq K \end{aligned} \quad (3b)$$

Finally, since we have finite capacity K , then:

$$P_n = 0 \quad \forall n \in n > K \quad (4)$$

Finally, from equations (1a, 3a, 4), final form:

$$P_n = \begin{cases} \frac{M!}{(M-n)!n!} \left(\frac{\lambda}{\mu}\right)^n P_0 & 0 \leq n \leq s \\ \frac{M!}{(M-n)!s!s^{n-s}} \left(\frac{\lambda}{\mu}\right)^n P_0 & s < n \leq K \\ 0 & n > K \end{cases} \quad (5a)$$

s.t: $s \leq K \leq M$

or equivalently, from equations (1b, 3b, 4) in binomial form:

$$P_n = \begin{cases} \binom{M}{n} \left(\frac{\lambda}{\mu}\right)^n P_0 & 0 \leq n \leq s \\ \frac{n!}{s!s^{n-s}} \binom{M}{n} \left(\frac{\lambda}{\mu}\right)^n P_0 & s < n \leq K \\ 0 & n > K \end{cases} \quad (5b)$$

s.t: $s \leq K \leq M$

(c)

Now assume that we have Y spare (redundant) servers on hand, such that:

- When a server fails or collapses, a spare is immediately substituted from the Y spare servers
- If all spares are already being used and another breakdown occurs, the system becomes short (i.e. available servers $< s$)
- When a failed server is repaired:
 - It becomes a spare again
 - Unless the system is short, in which case the repaired server immediately goes into service

Redo parts (a) and (b) above.

Constructing a 2-dimensional Markov Chain will be awarded with extra points.

To model this extension, we must introduce two additional parameters:

- α = failure rate of each currently available server
- β = repair rate of each failed server

The same number of customers may correspond to different numbers of available servers, so we'll use a 2D Markov chain for state n .

State definition

Let

$$(n, j)$$

where:

- n = number of customers in the system, $0 \leq n \leq K$
- j = number of failed servers, $0 \leq j \leq s + Y$

Since the system owns a total of $s + Y$ servers, the number of currently available servers is:

$$s + Y - j$$

However, at most s servers can be used simultaneously, so define the effective service capacity as:

$$c(j) = \min(s, s + Y - j)$$

Equivalently,

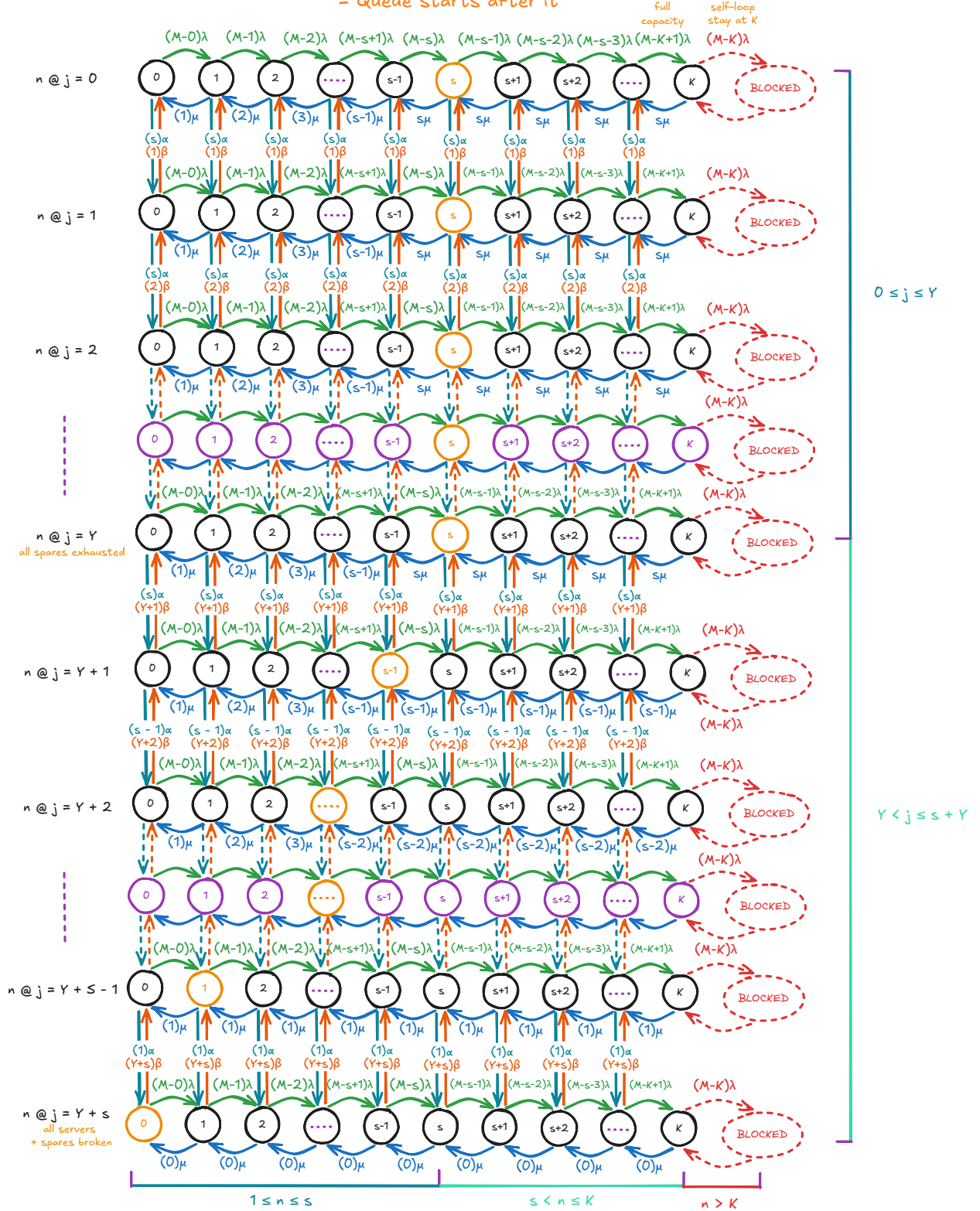
$$c(j) = \begin{cases} s & 0 \leq j \leq Y \\ s + Y - j & Y < j \leq s + Y \end{cases}$$

- If $j \leq Y$, all failed servers are fully covered by the spare pool, so the system still has s available servers.
- If $j > Y$, all spares are already consumed, so each additional failure reduces the number of available servers by 1.

(a) State-transition diagram

$M/M/s/K/M$
with Y spares and j broken servers

Orange node
= Queue starts after it



Transition rates

1. Arrivals

As before, only users outside the system can generate arrivals, so:

$$(n, j) \rightarrow (n + 1, j) \quad \text{with rate } (M - n)\lambda, \quad 0 \leq n < K$$

2. Service completions

The number of customers that can be served is limited by both the number of customers present and the number of available servers:

$$(n, j) \rightarrow (n - 1, j) \quad \text{with rate } \min(n, c(j))\mu, \quad 1 \leq n \leq K$$

3. Server failures

Only currently available servers can fail, so the failure transition is:

$$(n, j) \rightarrow (n, j + 1) \quad \text{with rate } c(j)\alpha, \quad 0 \leq j < s + Y$$

4. Server repairs

Each failed server is repaired independently, so:

$$(n, j) \rightarrow (n, j - 1) \quad \text{with rate } j\beta, \quad 1 \leq j \leq s + Y$$

- if the system is not short, the repaired server simply returns to the spare pool
- if the system is short, the repaired server immediately becomes active again

In both cases, the number of failed servers decreases by 1.

(b) Steady-state probabilities

Let

$$P_{n,j} = \Pr\{\text{system is in state } (n, j)\}$$

Then the process is a finite 2-dimensional continuous-time Markov chain, so the steady-state probabilities are obtained from the global balance equations plus normalization.

To write the balance equations compactly, recall that:

$$c(j) = \min(s, s + Y - j)$$

Global balance equations

1. Interior states

For

$$1 \leq n \leq K-1, \quad 1 \leq j \leq s+Y-1$$

all four transitions are possible, so:

$$\begin{aligned} & \left[(M-n)\lambda + \min(n, c(j))\mu + c(j)\alpha + j\beta \right] P_{n,j} \\ &= (M-n+1)\lambda P_{n-1,j} + \min(n+1, c(j))\mu P_{n+1,j} \\ & \quad + c(j-1)\alpha P_{n,j-1} + (j+1)\beta P_{n,j+1} \end{aligned}$$

Hence,

$$P_{n,j} = \frac{(M-n+1)\lambda P_{n-1,j} + \min(n+1, c(j))\mu P_{n+1,j} + c(j-1)\alpha P_{n,j-1} + (j+1)\beta P_{n,j+1}}{(M-n)\lambda + \min(n, c(j))\mu + c(j)\alpha + j\beta}$$

2. Bottom boundary: $j = 0$

For

$$1 \leq n \leq K-1$$

there is no repair transition from $(n, 0)$, hence:

$$\begin{aligned} & \left[(M-n)\lambda + \min(n, c(0))\mu + c(0)\alpha \right] P_{n,0} \\ &= (M-n+1)\lambda P_{n-1,0} + \min(n+1, c(0))\mu P_{n+1,0} \\ & \quad + \beta P_{n,1} \end{aligned}$$

Hence,

$$P_{n,0} = \frac{(M-n+1)\lambda P_{n-1,0} + \min(n+1, c(0))\mu P_{n+1,0} + \beta P_{n,1}}{(M-n)\lambda + \min(n, c(0))\mu + c(0)\alpha}$$

3. Top boundary: $j = s+Y$

For

$$1 \leq n \leq K-1$$

there is no further failure transition, hence:

$$\begin{aligned} & \left[(M-n)\lambda + \min(n, c(s+Y))\mu + (s+Y)\beta \right] P_{n,s+Y} \\ &= (M-n+1)\lambda P_{n-1,s+Y} + \min(n+1, c(s+Y))\mu P_{n+1,s+Y} \\ & \quad + c(s+Y-1)\alpha P_{n,s+Y-1} \end{aligned}$$

Hence,

$$P_{n,s+Y} = \frac{(M-n+1)\lambda P_{n-1,s+Y} + \min(n+1, c(s+Y))\mu P_{n+1,s+Y} + c(s+Y-1)\alpha P_{n,s+Y-1}}{(M-n)\lambda + \min(n, c(s+Y))\mu + (s+Y)\beta}$$

4. Left boundary: $n = 0$

For

$$1 \leq j \leq s + Y - 1$$

there is no service completion from $(0, j)$, hence:

$$\begin{aligned} & \left[M\lambda + c(j)\alpha + j\beta \right] P_{0,j} \\ &= \min(1, c(j))\mu P_{1,j} + c(j-1)\alpha P_{0,j-1} \\ & \quad + (j+1)\beta P_{0,j+1} \end{aligned}$$

Hence,

$$P_{0,j} = \frac{\min(1, c(j))\mu P_{1,j} + c(j-1)\alpha P_{0,j-1} + (j+1)\beta P_{0,j+1}}{M\lambda + c(j)\alpha + j\beta}$$

5. Right boundary: $n = K$

For

$$1 \leq j \leq s + Y - 1$$

there is no further arrival into the system, hence:

$$\begin{aligned} & \left[\min(K, c(j))\mu + c(j)\alpha + j\beta \right] P_{K,j} \\ &= (M - K + 1)\lambda P_{K-1,j} + c(j-1)\alpha P_{K,j-1} \\ & \quad + (j+1)\beta P_{K,j+1} \end{aligned}$$

Hence,

$$P_{K,j} = \frac{(M - K + 1)\lambda P_{K-1,j} + c(j-1)\alpha P_{K,j-1} + (j+1)\beta P_{K,j+1}}{\min(K, c(j))\mu + c(j)\alpha + j\beta}$$

6. Corner states

At the four corner states we keep only the valid transitions:

For $(0, 0)$:

$$\left[M\lambda + c(0)\alpha \right] P_{0,0} = \min(1, c(0))\mu P_{1,0} + \beta P_{0,1}$$

Hence,

$$P_{0,0} = \frac{\min(1, c(0))\mu P_{1,0} + \beta P_{0,1}}{M\lambda + c(0)\alpha}$$

For $(K, 0)$:

$$\left[\min(K, c(0))\mu + c(0)\alpha \right] P_{K,0} = (M - K + 1)\lambda P_{K-1,0} + \beta P_{K,1}$$

Hence,

$$P_{K,0} = \frac{(M - K + 1)\lambda P_{K-1,0} + \beta P_{K,1}}{\min(K, c(0))\mu + c(0)\alpha}$$

For $(0, s + Y)$:

$$\left[M\lambda + (s + Y)\beta \right] P_{0,s+Y} = \min(1, c(s + Y))\mu P_{1,s+Y} + c(s + Y - 1)\alpha P_{0,s+Y-1}$$

Hence ,

$$P_{0,s+Y} = \frac{\min(1, c(s + Y))\mu P_{1,s+Y} + c(s + Y - 1)\alpha P_{0,s+Y-1}}{M\lambda + (s + Y)\beta}$$

For $(K, s + Y)$:

$$\left[\min(K, c(s + Y))\mu + (s + Y)\beta \right] P_{K,s+Y} = (M - K + 1)\lambda P_{K-1,s+Y} + c(s + Y - 1)\alpha P_{K,s+Y-1}$$

Hence ,

$$P_{K,s+Y} = \frac{(M - K + 1)\lambda P_{K-1,s+Y} + c(s + Y - 1)\alpha P_{K,s+Y-1}}{\min(K, c(s + Y))\mu + (s + Y)\beta}$$

Collecting all these relations together:

$$P_{n,j} = \begin{cases} \frac{(M - n + 1)\lambda P_{n-1,j} + \min(n + 1, c(j))\mu P_{n+1,j} + c(j - 1)\alpha P_{n,j-1} + (j + 1)\beta P_{n,j+1}}{(M - n)\lambda + \min(n, c(j))\mu + c(j)\alpha + j\beta}, & 1 \leq n \leq K - 1, 1 \leq j \leq s \\ \frac{(M - n + 1)\lambda P_{n-1,0} + \min(n + 1, c(0))\mu P_{n+1,0} + \beta P_{n,1}}{(M - n)\lambda + \min(n, c(0))\mu + c(0)\alpha}, & 1 \leq n \leq K - 1, j = 0 \\ \frac{(M - n + 1)\lambda P_{n-1,s+Y} + \min(n + 1, c(s + Y))\mu P_{n+1,s+Y} + c(s + Y - 1)\alpha P_{n,s+Y-1}}{(M - n)\lambda + \min(n, c(s + Y))\mu + (s + Y)\beta}, & 1 \leq n \leq K - 1, j = s + Y \\ \frac{\min(1, c(j))\mu P_{1,j} + c(j - 1)\alpha P_{0,j-1} + (j + 1)\beta P_{0,j+1}}{M\lambda + c(j)\alpha + j\beta}, & n = 0, 1 \leq j \leq s \\ \frac{(M - K + 1)\lambda P_{K-1,j} + c(j - 1)\alpha P_{K,j-1} + (j + 1)\beta P_{K,j+1}}{\min(K, c(j))\mu + c(j)\alpha + j\beta}, & n = K, 1 \leq j \leq s \\ \frac{\min(1, c(0))\mu P_{1,0} + \beta P_{0,1}}{M\lambda + c(0)\alpha}, & (n, j) = (0, 0) \\ \frac{(M - K + 1)\lambda P_{K-1,0} + \beta P_{K,1}}{\min(K, c(0))\mu + c(0)\alpha}, & (n, j) = (K, 0) \\ \frac{\min(1, c(s + Y))\mu P_{1,s+Y} + c(s + Y - 1)\alpha P_{0,s+Y-1}}{M\lambda + (s + Y)\beta}, & (n, j) = (0, s + Y) \\ \frac{(M - K + 1)\lambda P_{K-1,s+Y} + c(s + Y - 1)\alpha P_{K,s+Y-1}}{\min(K, c(s + Y))\mu + (s + Y)\beta}, & (n, j) = (K, s + Y) \end{cases}$$

The normalization condition is:

$$\sum_{n=0}^K \sum_{j=0}^{s+Y} P_{n,j} = 1$$

Once all $P_{n,j}$ are solved, the probability of having n customers in the system is the marginal:

$$P_n = \sum_{j=0}^{s+Y} P_{n,j}$$

P_n is the marginal of the joint probability distribution $P_{n,j}$ with respect to the server-failure variable j .

Adding them all gives the total probability of just having n customers, regardless of server-failure status.\

Generator matrix form

Instead of writing all balance equations individually, we can define the infinitesimal generator matrix Q of the CTMC over the state space

$$\mathcal{S} = \{(n, j) : 0 \leq n \leq K, 0 \leq j \leq s + Y\}$$

For any two states $(n, j), (u, v) \in \mathcal{S}$, the entries of Q are:

$$Q((n, j), (u, v)) = \begin{cases} (M - n)\lambda, & (u, v) = (n + 1, j) \\ \min(n, c(j))\mu, & (u, v) = (n - 1, j) \\ c(j)\alpha, & (u, v) = (n, j + 1) \\ j\beta, & (u, v) = (n, j - 1) \\ -\left[\mathbf{1}_{\{n < K\}}(M - n)\lambda + \mathbf{1}_{\{n > 0\}}\min(n, c(j))\mu + \mathbf{1}_{\{j < s + Y\}}c(j)\alpha + \mathbf{1}_{\{j > 0\}}j\beta\right], & (u, v) = (n, j) \\ 0, & \text{otherwise} \end{cases}$$

So each row of Q contains:

- one entry for a possible arrival
- one entry for a possible service completion
- one entry for a possible server failure
- one entry for a possible repair
- one diagonal entry equal to the negative sum of all valid outgoing rates

All invalid transitions are automatically assigned rate 0, which is why the generator-matrix form removes the need to handle edge and corner states separately.

Why is the diagonal entry negative?

In a CTMC, the off-diagonal entries of a row give the rates of leaving the current state for other states. Therefore, the diagonal entry must be the negative of the total outgoing rate so that each row of Q sums to zero:

$$\sum_{(u, v) \in \mathcal{S}} Q((n, j), (u, v)) = 0$$

This is required by the generator-matrix definition and guarantees conservation of total probability.

Intuitively, over a very small interval Δt ,

$$\Pr\{(n, j) \rightarrow (u, v) \text{ in } \Delta t\} \approx Q((n, j), (u, v))\Delta t, \quad (u, v) \neq (n, j)$$

so the probability of remaining in the same state is approximately:

$$1 + Q((n, j), (n, j))\Delta t = 1 - \left(\text{total rate of leaving } (n, j)\right)\Delta t$$

Hence the diagonal must be negative.

If π is the steady-state row vector corresponding to the ordered states of Q , then:

$$\pi Q = 0, \quad \sum_{(n,j) \in \mathcal{S}} \pi_{n,j} = 1$$

Here, π is the row vector of all steady-state probabilities:

$$\pi = [\pi_{0,0}, \pi_{1,0}, \dots, \pi_{K,0}, \pi_{0,1}, \dots, \pi_{K,s+Y}]$$

with

$$\pi_{n,j} = P_{n,j}$$

The equation

$$\pi Q = 0$$

is just the matrix form of the global balance equations. It says that, at steady state, the probability distribution does not change with time, so the total probability flow into each state equals the total probability flow out of that state.

How to find the steady state from Q

After constructing Q , we solve for the steady-state row vector π from:

$$\pi Q = 0$$

subject to the normalization condition:

$$\sum_{(n,j) \in \mathcal{S}} \pi_{n,j} = 1$$

Procedure:

1. Fix an ordering of the state space $\mathcal{S}$, for example

$$(0,0), (1,0), \dots, (K,0), (0,1), \dots, (K,s+Y)$$

2. Construct the generator matrix Q according to that ordering.

3. Solve the linear system $\pi Q = 0$.

4. Since one equation is redundant, replace one of the equations by the normalization condition.

5. The resulting solution gives all probabilities $\pi_{n,j} = P_{n,j}$.

The reason one equation is redundant is that every row of Q sums to zero, so the equations in $\pi Q = 0$ are linearly dependent.

Small example with a 3×3 generator matrix:

Suppose a CTMC has three states 0,1,2 and generator matrix

$$Q = \begin{bmatrix} -2 & 2 & 0 \\ 1 & -4 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$

Let

$$\pi = [\pi_0, \pi_1, \pi_2]$$

Then

$$\pi Q = 0$$

gives:

$$\begin{aligned} -2\pi_0 + \pi_1 &= 0 \\ 2\pi_0 - 4\pi_1 + \pi_2 &= 0 \\ 3\pi_1 - \pi_2 &= 0 \end{aligned}$$

and normalization gives:

$$\pi_0 + \pi_1 + \pi_2 = 1$$

From the first and third equations:

$$\pi_1 = 2\pi_0, \quad \pi_2 = 3\pi_1 = 6\pi_0$$

Substitute into normalization:

$$\pi_0 + 2\pi_0 + 6\pi_0 = 9\pi_0 = 1$$

so

$$\pi_0 = \frac{1}{9}, \quad \pi_1 = \frac{2}{9}, \quad \pi_2 = \frac{6}{9} = \frac{2}{3}$$

Therefore,

$$\pi = \left[\frac{1}{9}, \frac{2}{9}, \frac{2}{3} \right]$$

The same idea is used for our queueing model, except that the state space is much larger and the entries of Q come from arrivals, service completions, failures, and repairs.

and the customer-count marginal is:

$$P_n = \sum_{j=0}^{s+Y} \pi_{n,j}$$

Final remark for part (c)

Unlike part (b), there is generally **no simple closed-form expression** for P_n in terms of only P_0 and the scalar parameters, because the system is no longer a 1-dimensional birth-death chain.

Therefore, the correct exact formulation is:

Form the balance equations or equivalently Q , solve for the steady state, then compute $P_n = \sum_{j=0}^{s+Y} P_{n,j}$

(d) Bonus

Submit a simulation of the given model, including:

- The simulation code
- The theoretical analysis
- Comparison results between theory and simulation

[Part (d) placeholder]